

OSCEOLA, IA, USA

WMO: 720701

Lat: 41.052N

Lon: 93.689W

Elev: 340

StdP: 97.31

Time Zone: -6.00 (NAC)

Period: 10-19

WBAN: 54942

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.8	-16.9	-23.7	0.5	-19.2	-21.2	0.6	-16.0	12.0	-1.3	11.1	-2.7	3.6	310	0.520

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	34.0	23.6	32.5	23.8	31.0	23.2	26.0	31.5	25.1	30.2	24.2	28.9	4.5	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.6	20.4	29.7	23.5	19.0	28.4	22.7	18.1	27.5	82.6	31.5	78.5	30.2	74.7	29.1	28.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.7	9.1	8.3	-26.2	36.2	2.6	1.4	-28.1	37.2	-29.6	38.0	-31.0	38.8	-32.9	39.8		
			-25.6	27.2	3.6	0.8	-28.2	27.8	-30.3	28.3	-32.3	28.8	-34.9	29.4		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.9	-4.2	-2.9	4.5	10.5	17.0	22.8	24.6	22.8	19.5	12.0	4.3	-1.0
	DBStd	11.46	7.02	7.20	6.65	5.35	5.10	3.35	3.31	2.84	4.78	5.28	6.04	6.26
	HDD10.0	1632	439	365	196	58	7	0	0	0	1	40	184	343
	HDD18.3	3324	697	596	430	238	87	4	2	2	43	205	421	600
	CDD10.0	1958	0	3	25	74	223	383	451	396	286	103	14	1
	CDD18.3	609	0	0	1	4	45	137	194	139	78	9	1	0
	CDH23.3	5568	0	0	12	53	381	1304	1968	1131	646	73	1	0
	CDH26.7	2047	0	0	1	8	108	488	836	373	221	12	0	0
(14) Wind	WSAvg	3.8	4.5	4.5	4.5	4.7	4.0	3.6	2.8	2.4	3.1	3.6	4.3	4.2
(15) Precipitation	PrecAvg	969	31	37	55	98	133	138	114	106	95	62	51	35
	PrecMax	1354	73	65	129	186	282	317	361	282	284	144	131	95
	PrecMin	675	15	8	0	40	35	20	45	16	31	4	0	4
	PrecStd	202	13	16	38	37	57	74	75	72	62	43	31	21
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.8	19.7	25.1	28.1	32.8	34.8	37.3	34.9	33.0	28.1	22.8	16.2
		MCWB	7.7	11.2	16.8	17.0	21.7	23.9	23.0	22.6	23.5	20.0	14.5	11.7
	2%	DB	10.1	14.4	21.2	25.0	29.8	32.9	34.9	32.0	31.5	25.4	18.6	12.0
		MCWB	6.6	10.4	13.8	16.2	21.0	23.7	24.4	24.0	23.1	17.3	12.9	8.7
	5%	DB	7.4	10.2	17.9	22.5	27.3	31.3	32.9	30.2	29.7	22.8	15.9	9.9
		MCWB	4.3	6.6	12.6	14.5	19.2	22.9	24.4	23.3	22.6	16.2	11.0	7.3
	10%	DB	5.0	7.0	14.8	20.0	25.2	29.7	31.2	28.7	27.3	20.8	13.2	7.3
		MCWB	2.9	4.2	10.4	13.4	18.5	22.5	23.9	22.5	21.0	15.4	8.9	4.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.1	13.9	17.6	19.4	23.4	26.5	27.3	26.7	25.1	21.8	15.9	13.1
		MCDB	10.9	17.1	24.3	25.7	30.0	32.8	33.0	31.5	31.1	25.2	19.8	15.2
	2%	WB	6.4	9.7	15.5	17.4	21.9	25.0	26.4	25.4	24.1	20.0	13.8	9.9
		MCDB	9.1	14.2	19.3	23.3	27.7	30.5	32.2	29.8	29.5	23.3	17.4	11.1
	5%	WB	4.7	7.0	12.9	15.9	20.8	24.1	25.6	24.3	23.2	18.2	11.8	7.1
		MCDB	7.3	9.8	16.7	21.1	25.8	29.4	30.9	28.4	27.5	21.2	15.4	9.3
	10%	WB	2.8	4.1	10.5	14.3	19.8	23.2	24.6	23.3	22.1	16.0	9.4	4.8
		MCDB	5.0	6.7	14.7	18.5	24.0	28.5	29.2	27.3	26.0	19.7	13.2	7.2
(35) Mean Daily Temperature Range	5% DB	MDBR	9.7	9.7	10.1	11.5	10.8	10.9	10.9	10.6	11.3	11.5	10.4	8.5
		MCDBR	12.0	14.0	15.1	15.8	13.0	12.2	12.1	11.9	12.3	13.5	14.0	11.4
	5% WB	MCWBR	8.7	9.4	8.7	8.6	6.1	5.2	4.7	5.5	5.3	7.0	8.5	8.1
		MCWBR	11.5	13.7	12.5	13.7	11.2	10.8	10.4	10.5	10.7	10.7	12.5	10.1
(40) Clear-Sky Solar Irradiance	taub		0.288	0.318	0.330	0.386	0.409	0.420	0.432	0.421	0.380	0.339	0.298	0.285
		taud	2.428	2.345	2.399	2.260	2.264	2.274	2.237	2.286	2.352	2.463	2.510	2.459
	Ebn at Noon		873	890	921	885	869	857	842	842	855	859	857	847
		Edh at Noon	78	100	106	131	134	133	137	126	110	88	72	71
(44) All-Sky Solar Radiation	RadAvg		1.96	2.92	3.78	4.79	5.64	6.27	6.31	5.48	4.65	3.14	2.15	1.66
	RadStd		0.19	0.21	0.33	0.34	0.45	0.59	0.47	0.34	0.33	0.36	0.18	0.16

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (47 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page